Teaching LLMs Brazilian Healthcare: Injecting Knowledge from Official Clinical Guidelines

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Abstract. Brazil's Unified Health System (SUS) relies on official clinical guidelines that govern diagnostic criteria, treatments, and monitoring procedures for over 200 million citizens. Current Large Language Models (LLMs)—even state-of-the-art proprietary ones—perform poorly when probed on this guideline-specific knowledge, and, to the best of our knowledge, no prior work has attempted to inject it into models' parametric knowledge, nor does any benchmark exist to evaluate guideline-grounded clinical recall in Brazilian Portuguese. We address both gaps with a reproducible training protocol that adapts Qwen2.5-14B-Instruct to the Brazilian clinical domain. Starting from the full set of official guidelines (∼5.4M tokens), we generate approximately 70 million tokens of synthetic training data in three complementary formats—rephrases, wiki-style articles, and question-answer pairs—using four diverse generator LLMs to maximize stylistic and structural variety. The resulting corpus is used for continual pre-training, followed by reinforcement learning via Group Relative Policy Optimization (GRPO) to encourage explicit clinical reasoning. We introduce two protocol-grounded benchmarks: HealthBench-BR, comprising 1,780 balanced true/false clinical assertions, and PCDT-QA, containing 890 open-ended clinical questions scored by an LLM judge. Our best configuration achieves 83.9% on HealthBench-BR and 85.4% on PCDT-QA, outperforming GPT-5.2 despite being a 14B-parameter open model. Ablations confirm that both multi-generator diversity and reinforcement learning are critical to the observed gains. We publicly release all datasets, benchmarks, and model weights to support reproducible research on clinical NLP for Brazilian Portuguese.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

Brazil's Unified Health System (SUS – *Sistema Único de Saúde*) provides universal healthcare coverage to over 200 million people, making it one of the largest public health systems in the world. Clinical decision-making within SUS is governed by official clinical guidelines published by the Ministry of Health, which

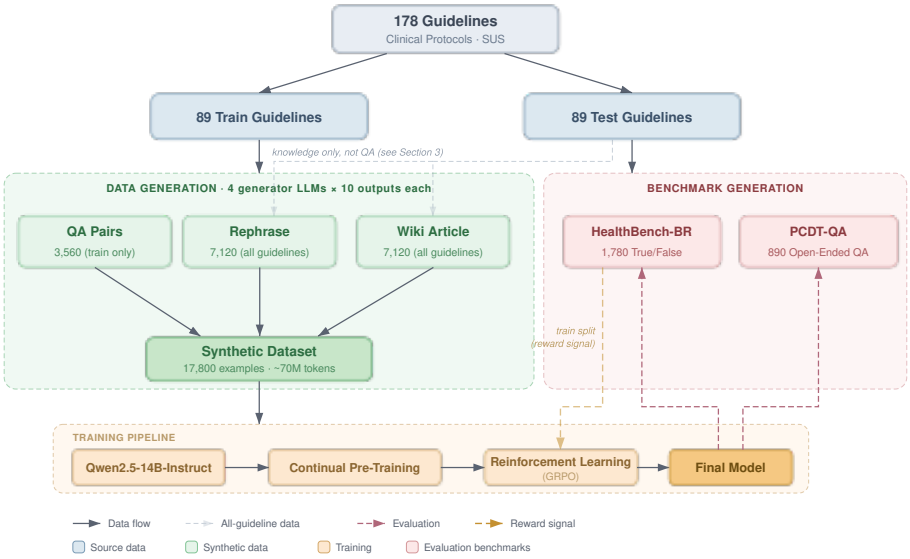

Fig. 1. Overview of our pipeline. The 178 clinical guidelines are split into train and test sets. All guidelines feed rephrase and wiki-style generation (dashed gray arrows), while QA pairs are generated only from train guidelines to prevent leakage. Four generator LLMs produce the synthetic corpus (∼70M tokens), which is used for continual pre-training of Qwen2.5-14B-Instruct. The HealthBench-BR train split provides the reward signal for GRPO reinforcement learning (dashed amber arrow). Both benchmarks are evaluated on the final model (dashed red arrows).

prescribe diagnostic criteria, treatment regimens, dosages, and monitoring procedures for conditions treated under public care [4]. These documents—primarily Clinical Protocols and Therapeutic Guidelines (PCDTs), but also several related categories described in Section 3—are hosted under the Ministry's PCDT portal.[4]. Adherence to these protocols is mandatory for SUS managers, yet their volume and complexity make it difficult for healthcare professionals to recall specific recommendations at the point of care. Large language models (LLMs) offer a promising path toward intelligent clinical decision support, yet current medical LLMs are predominantly trained on English-language medical literature and evaluated on English benchmarks, leaving their knowledge of Brazilian clinical guidelines largely untested [5,11].

Even among multilingual medical LLMs, Portuguese remains underrepresented. BioMistral [15] was trained exclusively on English text from PubMed Central; its multilingual evaluation—the first at scale for medical LLMs—relies on machine-translated benchmarks, offering no guarantee that the model has internalized the clinical standards of the Brazilian public health system. MMed-

[4] `https://www.gov.br/saude/pt-br/assuntos/pcdt`

Llama 3 [22] constructed a six-language medical corpus of 25.5B tokens but excludes Portuguese from both training data and evaluation. Within Portuguese medical NLP, prior work has addressed clinical NER with encoder models [25], parameter-efficient fine-tuning on clinical notes [21], and evaluation on residency entrance exams [19,9]—but no study has attempted to inject the knowledge contained in official Brazilian clinical protocols into the parameters of an LLM, and no benchmark exists to evaluate whether a model can accurately recall guideline-grounded clinical knowledge.

We address both gaps by proposing a training protocol for adapting a general-purpose LLM to the clinical domain of the Brazilian public health system. Starting from all 178 clinical guidelines described above, we first generate approximately 70 million tokens of synthetic training data across three complementary formats (rephrases, wiki-style articles, and question-answer pairs) using four diverse generator LLMs, following multi-style rephrasing strategies shown to be effective for domain adaptation [17,7]. This corpus is used for continual pre-training (CPT) of Qwen2.5-14B-Instruct [34], followed by reinforcement learning via Group Relative Policy Optimization (GRPO) [27] to encourage explicit clinical reasoning. To evaluate knowledge absorption, we introduce two guideline-grounded benchmarks: **HealthBench-BR**, comprising 1,780 balanced true/false clinical assertions requiring precise factual recall, and **PCDT-QA**, containing 890 open-ended clinical questions scored by an LLM judge.

Our best configuration outperforms GPT-5.2 on both benchmarks despite being a 14B-parameter open model. In summary, our main contributions are: (i) a reproducible training protocol combining multi-generator synthetic data, continual pre-training, and reinforcement learning for clinical domain adaptation; (ii) two protocol-grounded evaluation benchmarks for the Brazilian public health system; (iii) a detailed analysis of the trade-offs between knowledge injection and catastrophic forgetting, including ablations showing that both multi-generator diversity and reinforcement learning are critical to the observed gains; and (iv) the public release of all synthetic datasets, benchmarks, and model weights to support reproducible research on clinical NLP for Brazilian Portuguese.[5]

2 Related Work

2.1 Medical LLMs

Proprietary models have driven rapid progress on medical benchmarks: Med-PaLM [29] first surpassed the USMLE passing threshold, Med-PaLM 2 [30] reached 86.5%, Med-Gemini [24] pushed to 91.1%, and GPT-4 exceeded it by over 20 points without medical prompting [18]. However, their closed nature and English-centric design limit applicability in settings like Brazilian public health, where transparency, reproducibility, and native-language reasoning are essential.

On the open-source side, continual pre-training (CPT) has emerged as the dominant adaptation strategy. MEDITRON [6] adapted Llama-2 on 48B to-

[5] URL anonymized for review.

kens including clinical guidelines with GAP-Replay, PMC-LLaMA [32] and Me-LLaMA [33] scaled CPT on biomedical literature to 129B tokens, and Christophe et al. [8] showed that CPT on clinical text provides a strong foundation for subsequent instruction tuning. Multilingual efforts remain scarce: BioMistral [15] conducted the first large-scale multilingual medical evaluation—including Portuguese via automatic translation—and MMed-Llama 3 [22] built a six-language corpus that excludes Portuguese entirely. Critically, neither incorporates official clinical protocols as a knowledge source.

2.2 Portuguese Medical NLP

Early work focused on encoder models, such as BioBERTpt [25], which adapted BERT for clinical NER. Clinical-BR-LlaMA/Mistral [21] applied LoRA on clinical notes, and DrBode [23] fine-tuned ChatBode-7B on translated medical data with QLoRA, though adapted models underperformed the base due to catastrophic forgetting. On the data and evaluation side, MedPT [10] released 384K native patient–doctor QA pairs, studies have benchmarked LLMs on the Revalida exam [19], and HealthQA-BR [9] provides 5,632 residency questions—yet none evaluates factual recall over official Brazilian clinical protocols, and no prior work injects parametric knowledge from these protocols into LLMs. We address both gaps: we adapt an LLM on 178 PCDTs through continual pre-training and reinforcement learning, and introduce two protocol-grounded benchmarks—HealthBench-BR (true/false) and PCDT-QA (open-ended)—to measure knowledge absorption (Section 5).

2.3 Synthetic Data for Domain Adaptation

Training on high-quality synthetic data can match or exceed training on much larger natural corpora. Gunasekar et al. [12] demonstrated that models trained primarily on synthetic textbook-quality data achieve strong coding performance at a fraction of the usual data scale. WRAP [17] generalized this insight to pre-training: by prompting an instruction-tuned model to rephrase web documents in multiple styles—Wikipedia-like, question–answer, and paraphrase—it speeds up pre-training while improving downstream accuracy. Nemotron-CC [31] scaled this source-rephrasing approach to trillions of tokens, and AdaptLLM [7] showed that reformatting raw domain corpora as reading comprehension tasks enables small models to rival much larger domain-specific ones. In medicine, AlpaCare [36] and ChatDoctor [16] demonstrated that LLM-generated instructions and dialogues substantially boost clinical performance from limited seeds. Our synthetic data pipeline follows this philosophy, transforming 178 official Brazilian protocols into rephrases, wiki-style articles, and QA pairs—similar to the multi-style strategy of WRAP but applied to a small, high-value domain corpus to maximize knowledge absorption.

2.4 Catastrophic Forgetting in Continual Pre-Training

A key risk in CPT is catastrophic forgetting, which scales as a power law in parameters and update steps [14]. Parameter-efficient methods like LoRA naturally mitigate this by modifying fewer parameters [13,3], and replay of even 1% of pre-training data can prevent it [2]—though recent work suggests that some observed drops reflect alignment loss rather than true knowledge erasure [37]. We opt for full CPT to maximize knowledge injection from our small domain corpus, explicitly measure forgetting on general benchmarks, and provide ablations comparing the trade-offs of forgetting-mitigation strategies.

3 Clinical Guideline Corpus

The Brazilian Ministry of Health publishes a set of official clinical guidelines for the Unified Health System (SUS) that establish diagnostic criteria, treatment regimens, dosages, and monitoring procedures for specific diseases and health conditions [4]. These documents fall into several regulatory categories: Clinical Protocols and Therapeutic Guidelines (PCDTs – *Protocolos Clínicos e Diretrizes Terapêuticas*), which form the majority; Protocols of Use (*Protocolos de Uso*); Oncology Diagnostic and Therapeutic Guidelines (DDTs – *Diretrizes Diagnósticas e Terapêuticas em Oncologia*); National Guidelines (*Diretrizes Brasileiras*); and Care Pathways (*Linhas de Cuidado*). All are elaborated with support from the National Commission for the Incorporation of Technologies (CONITEC – *Comissão Nacional de Incorporação de Tecnologias no SUS*), and their adoption is mandatory for SUS managers. For brevity, we refer to the full collection as *clinical guidelines* throughout this paper. We collected all 178 guidelines available as of March 2026 as PDF documents from the Ministry of Health portal.[6]

We collected all 178 clinical guidelines available as of March 2026 as PDF documents from the Ministry of Health portal.[7] The text content was extracted from each PDF, yielding a raw corpus of approximately 26.9 million characters. Individual protocols range from 18k to over 915k characters (median 117k). To fit within the context window of our generator models (Section 4), we truncate each protocol at 120k characters, retaining the full text for the majority of protocols and the initial sections—which contain diagnostic criteria, treatment recommendations, and monitoring parameters—for longer ones. After truncation, the corpus comprises approximately 16.6M characters (5.4M tokens). The extracted protocols and all derived datasets are publicly available.[8]

4 Synthetic Data Pipeline

Rather than training on the original guidelines, we use LLMs to transform each guideline into synthetic texts in three complementary formats (Figure 1, green

[6] `https://www.gov.br/saude/pt-br/assuntos/pcdt`
[7] `https://www.gov.br/saude/pt-br/assuntos/pcdt`
[8] URL anonymized for review.

blocks), following recent work showing that rephrasing corpora into diverse formats improves knowledge absorption during pre-training [17,7]. Each guideline is processed under three distinct prompts, producing 10 outputs per prompt:

Rephrase. The generator rewrites the guideline, modifying information order and paraphrasing while preserving all clinical content. This exposes the model to the same knowledge in varied surface forms. Rephrases are generated for all 178 guidelines.

Wiki-style article. The generator produces an expository article in the style of Wikipedia, restructuring the prescriptive clinical document while retaining all key numerical values and details. Wiki articles are generated for all 178 guidelines.

Question–answer pairs. The generator creates questions and detailed answers grounded in the guideline, ranging from broad clinical queries to specific dosage and monitoring questions. Answers include step-by-step rationales to encourage internalization of clinical reasoning patterns. To prevent data leakage, QA pairs are generated only from the 89 train-split guidelines, as illustrated by the dashed arrows in Figure 1: test-split guidelines feed only rephrase and wiki generation, never QA.

To increase the stylistic and lexical diversity of the synthetic corpus, each of the three generation strategies is applied independently by four distinct models: GPT-4.1-mini, GPT-5-nano, GPT-OSS-20B, and Qwen3-235B. As shown in Table 2, scaling from one to four generators substantially improves downstream performance. We also evaluate self-generation with the target model (Qwen2.5-14B-Instruct), which yields the weakest results; we analyze this case in Section 7.3.

The full synthetic corpus comprises 17,800 examples (70M tokens), a roughly 13× expansion over the original 5.4M-token corpus with substantially greater format and stylistic diversity. This dataset is used for continual pre-training (Section 6).

5 Evaluation Benchmarks

We introduce two complementary benchmarks derived from the clinical guidelines. Both are split into training and test by guideline, with 89 guidelines in each split. Questions are generated for all 178 guidelines, but we report accuracy separately on train and test splits; performance on test-split questions reflects generalization to guidelines seen only as rephrase and wiki data during training, never as QA supervision.

5.1 HealthBench-BR

HealthBench-BR is a true/false benchmark comprising 1,780 clinical assertions (10 per guideline, 890 per split). Assertions are generated in paired form: for

each clinical fact, one statement is faithful to the guideline (true) and a second modifies a key detail—such as a dosage, route of administration, or monitoring interval—to produce a plausible but false variant. This yields a perfectly balanced dataset (50% true, 50% false) and ensures that correct classification requires precise factual recall rather than surface-level heuristics. During evaluation, the model is prompted to conclude with an explicit verdict (*Verdadeiro*/*Falso*), extracted via regex. All inference uses greedy decoding.

5.2 PCDT-QA

PCDT-QA is an open-ended benchmark containing 890 questions (5 per guideline, 445 per split) covering both broad clinical topics (e.g., inclusion criteria) and specific factual queries (e.g., recommended pediatric dosages). Each question is paired with a reference answer grounded in the guideline text. Evaluation uses an LLM-as-a-judge pipeline: the model under evaluation generates a free-text answer (greedy decoding), and GPT-4.1 compares it against the reference, producing a binary verdict (correct/incorrect). This accommodates the open-ended nature of clinical responses, where correct answers may differ substantially in wording from the reference.

6 Experimental Setup

6.1 Continual Pre-Training

We perform continual pre-training (CPT) on Qwen2.5-14B-Instruct [34] using a standard causal language modeling objective over the full synthetic corpus. Training runs on a TPU v6-64 pod using JAX. We train for 3 epochs with a cosine learning rate schedule (peak 5×10^{-5}, minimum 10% of peak) and AdamW (β_1=0.9, β_2=0.95). The effective batch size is 32. Sequences are truncated to 4,096 tokens; all tokens contribute to the loss. We use a warmup ratio of 1/3 of total steps.

6.2 Reinforcement Learning with GRPO

After CPT, we apply Group Relative Policy Optimization (GRPO) [27] on the HealthBench-BR train split to encourage explicit clinical reasoning over true/false assertions. Rather than memorizing surface-level patterns, the model must evaluate factual clinical claims—e.g., whether a given dosage is appropriate for a specific condition—and produce a justified verdict. This verifiable structure makes the task well-suited to reinforcement learning with verifiable rewards (RLVR), where correctness can be checked automatically.

Recent work has shown that parameter-efficient methods are sufficient for RLVR: Schulman et al. [26] demonstrated that LoRA matches full fine-tuning for RL even at very low ranks, since policy gradient methods absorb far less information per episode than supervised learning, and Yin et al. [35] confirmed

that PEFT methods are broadly effective under the RLVR paradigm. Accordingly, we train GRPO with LoRA (r=32, α=64, applied to all linear layers) for 5 epochs, generating 16 completions per prompt with a maximum length of 512 tokens. The effective batch size is 128 and the KL penalty coefficient β is set to 0 (no explicit KL constraint), relying on the clipping parameters (ϵ=0.2, ϵ_{high}=0.28) to bound policy updates.

The reward function assigns 1.0 to correct answers that contain at least 50 words of reasoning, and 0.0 otherwise. This minimum-length requirement serves a dual purpose: it prevents reward hacking, where the model could learn to emit bare "Verdadeiro"/"Falso" labels without justification, and it incentivizes the model to develop clinical reasoning—verbalizing relevant diagnostic criteria, contraindications, or dosage considerations before reaching a verdict.

7 Results

7.1 Main Results

Table 1 summarizes performance across all configurations and baselines. The base Qwen2.5-14B-Instruct model, without any domain-specific training, achieves 59.4% on HealthBench-BR and 27.9% on PCDT-QA (test split), indicating that guideline-specific knowledge is largely absent from the model's parameters.

Continual pre-training with a single generator raises test performance to 69.6% on HealthBench-BR and 66.3% on PCDT-QA—already approaching GPT-4.1 on both benchmarks. Scaling to four generators further improves PCDT-QA to 86.3%, substantially surpassing all frontier models, while HealthBench-BR reaches 71.1%. The disproportionate gain on PCDT-QA suggests that generator diversity primarily benefits free-form knowledge recall, where exposure to varied phrasings and explanatory styles translates directly into richer answers.

Following CPT with GRPO yields the strongest HealthBench-BR results: 83.9% on the test split with four generators, a 12.8 percentage point gain over CPT alone. The effect on PCDT-QA is more nuanced: GRPO produces a small decrease from 86.3% to 85.4%, likely because the reward objective optimizes specifically for true/false clinical verification rather than open-ended generation. Despite this marginal trade-off, the final model remains above all frontier baselines on both benchmarks.

Among frontier models, Gemini 3.1 Pro is the strongest baseline at 79.3% on HealthBench-BR and 80.0% on PCDT-QA, followed closely by GPT-5.2 (high). Our best configuration outperforms Gemini 3.1 Pro by 4.6 and 5.4 percentage points, respectively, despite being a 14B-parameter open model. Notably, even our CPT-only configuration (4 generators, no GRPO) already surpasses all frontier models on PCDT-QA by over 6 points, suggesting that continual pre-training on guideline-specific synthetic data is highly effective for knowledge absorption in this domain.

We also compare against Google AI Overview,[9] which generates answers grounded in web search results—effectively a retrieval-augmented generation

[9] `https://blog.google/products/search/generative-ai-google-search-may-2024`

Table 1. Main results on HealthBench-BR (true/false accuracy) and PCDT-QA (open-ended accuracy via LLM judge). Train and Test refer to the guideline split (Section 5). All values in %. Baseline is Qwen2.5-14B-Instruct without domain-specific training.

	HealthBench-BR		**PCDT-QA**	
Model	**Train**	**Test**	**Train**	**Test**
Ours				
Baseline (no training)	61.1	59.4	30.8	27.9
+ RL only	68.7	64.2	31.5	29.4
+ SFT only	74.0	62.1	32.8	29.4
+ CPT	67.1	69.6	73.5	66.3
+ CPT + SFT	83.5	70.4	64.9	57.8
+ CPT + RL	85.5	75.6	70.6	65.4
+ CPT (4 generators)	70.9	71.1	91.0	86.3
+ CPT (4 generators) + SFT	87.0	76.4	87.6	81.1
+ CPT (4 generators) + RL	90.4	83.9	88.8	85.4
Frontier models				
GPT-4.1	73.1	73.3	70.1	70.3
GPT-5.2 (instant)	72.5	76.3	69.2	71.7
GPT-5.2 (high)	78.9	78.5	74.6	78.2
Claude Sonnet 4.6	77.2	77.6	67.0	70.3
Gemini 3.1 Pro	77.8	79.3	74.8	80.0
RAG				
Google AI Overview	72.4	70.5	77.1	77.3
Baseline + RAG (BM25)	88.4	90.0	56.0	58.2
Baseline + RAG (embeddings)	86.5	87.9	83.6	86.7
CPT + RAG (BM25)	87.4	89.0	79.8	78.9
CPT + RAG (embeddings)	86.1	85.2	91.2	92.1
CPT (4 gen) + RAG (BM25)	74.2	73.9	92.1	88.3
CPT (4 gen) + RAG (embeddings)	74.0	71.1	92.1	94.8
CPT (4 gen) + RL + RAG (BM25)	94.9	95.3	90.6	88.8
CPT (4 gen) + RL + RAG (embeddings)	94.2	94.0	92.6	93.3

(RAG) system—and is arguably one of the most widely deployed sources of AI-generated clinical information, given its integration into Google Search. For evaluation, we submit each benchmark question as a search query and extract the generated summary when available. Notably, AI Overview does not produce a response for every query: it abstained on 11.7% of HealthBench-BR and 4.9% of PCDT-QA questions, and the reported accuracy considers only questions where a response was generated, giving it an inherent advantage. Even under this favorable comparison, our best configuration outperforms AI Overview by 13.4 points on HealthBench-BR and 8.1 points on PCDT-QA, demonstrating that targeted knowledge injection into a 14B model can surpass retrieval-augmented access to the open web.

Table 2. Effect of generation format and generator diversity on CPT performance (test split). HB = HealthBench-BR, QA = PCDT-QA. "1 generator" uses GPT-4.1-mini only; "4 generators" uses all four models described in Section 4.

	1 generator			4 generators		
Generation format	#	**HB**	**QA**	#	**HB**	**QA**
Rephrase	1,780	62.8	52.4	7,120	63.6	73.0
Wiki	1,780	64.2	60.9	7,120	69.7	76.9
Rephrase + QA	2,670	63.7	56.6	10,680	66.3	75.3
Wiki + QA	2,670	67.0	62.2	10,680	68.2	76.0
Wiki + Rephrase	3,560	63.0	65.4	14,240	68.4	83.8
Rephrase + Wiki + QA	4,450	69.6	66.3	17,800	71.1	86.3

An important dimension beyond accuracy is sycophancy—the tendency to agree with the user regardless of factual correctness. On HealthBench-BR, a well-calibrated model should predict "True" for approximately 50% of questions, since the dataset is perfectly balanced. The base Qwen2.5-14B-Instruct is well calibrated at 49.9%, but CPT shifts this to 58.9%, suggesting that continual pre-training on synthetic data introduces a bias toward agreement, possibly because the training corpus tends toward affirmative framing. GRPO effectively corrects this: our best configuration (CPT + RL, 4 generators) returns to 53.0%, indicating that the reinforcement learning stage, by requiring the model to justify its verdict, teaches it to disagree when the evidence warrants it. Among frontier models, Claude Sonnet 4.6 (49.2%) and GPT-5.2 high (48.7%) are well calibrated, while Gemini 3.1 Pro exhibits a moderate agreement bias at 63.3%—noteworthy given that it is the strongest frontier baseline on both benchmarks. The most concerning case is Google AI Overview, which predicts "True" 69.6% of the time, making it the most sycophantic system evaluated. Given its reach as the default AI-generated response in Google Search, this tendency to confirm clinical assertions—including false ones about dosages, contraindications, and monitoring intervals—raises concerns for users seeking clinical information without professional guidance.

7.2 Effect of Generation Formats

Table 2 isolates the contribution of each augmentation format under both single- and multi-generator settings. Across all configurations, wiki-style articles consistently outperform rephrases—for instance, with a single generator, wiki alone reaches 64.2% on HealthBench-BR and 60.9% on PCDT-QA, compared to 62.8% and 52.4% for rephrase. This advantage likely reflects the expository nature of the wiki format, which restructures protocol content into self-contained explanations that more closely resemble the reasoning required at evaluation time.

Combining formats is consistently beneficial: each additional format improves at least one benchmark, and the full combination (rephrase + wiki + QA) yields the best results in every setting. With four generators, adding QA to the wiki

Table 3. Effect of the generator model on CPT performance. Each model generates all three augmentation types (rephrase, wiki, QA) for 4,450 training examples.

Generator model	Tokens	Avg. doc	HB (%)	OE (%)
Qwen2.5-14B (self-aug)	7.1M	1,594	60.2	50.1
GPT-OSS-20B	15.4M	3,471	65.7	64.0
GPT-5-nano	15.7M	3,522	65.2	67.4
GPT-4.1-mini	18.1M	4,065	69.6	66.3
Qwen3-235B	20.3M	4,551	63.5	66.7

+ rephrase combination raises HealthBench-BR from 68.4% to 71.1% (+2.7 points) and PCDT-QA from 83.8% to 86.3% (+2.5 points), confirming that direct question–answer supervision provides a complementary learning signal that document-style formats alone do not fully capture.

7.3 Effect of the Generator

Table 3 compares the five generator models in isolation, each producing all three generation formats. The clear outlier is self-generation with the base model itself (Qwen2.5-14B-Instruct), which yields the weakest results by a wide margin—60.2% on HealthBench-BR and 50.1% on PCDT-QA. We attribute this to output length: Qwen2.5-14B averages 1,594 tokens per example against 3,471–4,551 for the external generators, producing only 7.1M tokens total. Shorter outputs inevitably omit the specific dosages, monitoring intervals, and inclusion criteria that are central to both benchmarks.

Among the four external generators, no single model leads on both benchmarks: GPT-4.1-mini achieves the best HealthBench-BR score (69.6%), while GPT-5-nano leads on PCDT-QA (67.4%). Notably, Qwen3-235B does not dominate either benchmark despite producing the most tokens, suggesting diminishing returns from additional generator capacity.

This motivates combining all four external generators rather than relying on the strongest one. Generator and format diversity are complementary: with one generator, all three formats combined yield 66.3% on PCDT-QA; scaling to four generators on the same combination reaches 86.3% (Table 2)—a 20.0 point gain that far exceeds any single-format improvement.

7.4 Catastrophic Forgetting Analysis

Table 4 reports performance on five out-of-distribution (OOD) benchmarks alongside our two in-distribution evaluations, comparing two forgetting-mitigation strategies. The OOD set spans HealthQA-BR [9], DrBode [23], MedPT [10], BLUEX [1] (Brazilian university admissions, general PT knowledge), and IFEval [38] (instruction-following). We compare full fine-tuning, LoRA, and full FT with 50% replay drawn from FineWeb [20] as an English-language proxy for the Qwen2.5 pre-training distribution.

Table 4. Comparison of catastrophic forgetting-mitigation strategies.

		In-distribution		Out-of-distribution					
Data	**Method**	HB	PCDT	HealthQA	DrBode	MedPT	BLUEX	IFEval	Avg
Baseline	–	59.4	27.9	67.8	65.3	56.4	75.0	79.7	68.8
1 gen	Full FT	69.6	66.3	66.0	62.7	56.4	73.5	75.8	66.9
	LoRA	67.2	65.2	64.1	56.6	53.8	71.1	68.6	62.8
	Replay	63.8	65.6	67.0	64.5	55.9	72.7	77.1	67.4
4 gen	Full FT	71.1	86.3	63.3	61.2	54.1	72.8	69.3	64.1
	LoRA	70.3	77.1	60.8	58.3	52.1	68.9	65.6	61.1
	Replay	–	–	–	–	–	–	–	–

Full fine-tuning yields moderate forgetting: -1.9 points on average with one generator and -4.7 with four. The larger drop tracks the larger in-distribution gain (PCDT-QA 66.3 $\rightarrow$ 86.3), consistent with the power-law trade-off in [14]. IFEval is the most affected (-4 to -10 points), suggesting that prolonged optimization on document-style text partially erodes instruction-following alignment.

Counterintuitively, LoRA is dominated by full FT on both axes—more OOD forgetting (-6.0 vs. -1.9 with one generator; -7.7 vs. -4.7 with four) and lower in-distribution scores (PCDT-QA 77.1 vs. 86.3). This contrasts with [3], who report that LoRA both learns and forgets less than full FT, possibly because our setting differs in corpus size, stylistic homogeneity, and rank (r=32). The pattern is instead consistent with the *intruder dimensions* account of [28], in which LoRA solutions contain high-magnitude singular vectors approximately orthogonal to the base model—absent from full-FT solutions—that degrade unrelated tasks; this matches our largest gap being on IFEval (-11.1 and -14.1 vs. $-3.9/-10.4$), where these directions plausibly overlap with instruction-following parameters.

Mixing FineWeb at a 1:1 ratio recovers the OOD average to within 1.4 points of the baseline (67.4 vs. 68.8) while retaining most of the in-distribution gain (HealthBench-BR 63.8, PCDT-QA 65.6), consistent with [2]. The cost is $\sim$6 points on HealthBench-BR relative to full FT without replay. Since FineWeb is predominantly English, part of the retention—especially on IFEval—may reflect continued English exposure rather than genuine preservation of Portuguese capabilities; a PT replay source is a natural follow-up.

Overall, full FT sits on the favorable side of the knowledge–retention frontier when OOD performance is secondary; replay is safer when general capability retention matters; LoRA is dominated on both axes. We adopt full FT for the main results.

8 Ethical Considerations

Our model is designed to support, not replace, clinical judgment: even at 83.9% on HealthBench-BR, roughly one in six assertions is misclassified, and residual

errors may involve consequential details such as dosages or contraindications. The released artifact is a research model, not a certified medical device, and should be deployed only within workflows supervised by licensed professionals. A related concern is sycophancy: Google AI Overview confirms true/false assertions 69.6% of the time on a balanced benchmark, a bias we show GRPO with reasoning-length rewards substantially corrects. All training data derives from public Ministry of Health guidelines, avoiding privacy risks inherent to clinical notes, though residual biases from the source documents and generator models may persist. Finally, open release of models trained on Brazilian protocols is a step toward equitable access for Portuguese-speaking clinicians and patients underserved by English-centric medical LLMs.

9 Conclusion

We presented a reproducible protocol for adapting an open LLM to the Brazilian Unified Health System, using 178 official Ministry of Health guidelines as the sole authoritative source. Our pipeline expands a 5.4M-token corpus into approximately 70M tokens of synthetic data across three formats and four generators; continual pre-training of Qwen2.5-14B-Instruct followed by GRPO yields 83.9% on HealthBench-BR and 85.4% on PCDT-QA, outperforming GPT-5.2, Claude Sonnet 4.6, and Gemini 3.1 Pro despite being a 14B open model. Ablations show that generator and format diversity are complementary—scaling to four generators on the full three-format corpus adds 20 points on PCDT-QA—while GRPO is critical for verdict-style recall (+12.8 points on HealthBench-BR) and simultaneously corrects post-CPT sycophancy. Limitations include evaluation grounded in guideline recall rather than real clinical encounters, the need for periodic retraining as protocols evolve, and prospective validation with healthcare professionals before any deployment. We release all datasets, benchmarks, and weights to support open, language-native medical assistants for underrepresented healthcare systems.

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